

MATA37H3 — Week 01

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Sigma Notation, Riemann Sums & the Definite Integral)

EXAMPLE 1 — Answer

Both methods give $\sum_{i=1}^6 (2i - 3) = 2(21) - 3(6) = 42 - 18 = 24$

Common misconception: Treating $\sum (2i - 3)$ as $2 \sum i - 3$ (forgetting the -3 must also be summed n times, i.e. it's $-3n$, not just -3).

EXAMPLE 2 — Answer

$$\sum_{i=1}^{40} i^3 = \frac{40^2(41)^2}{4} = 672,400$$

Common misconception: Using the formula for $\sum i^2$ instead of $\sum i^3$, or forgetting to square the numerator $n(n + 1)$ before dividing by 4.

EXAMPLE 3 — Answer

$$\text{Telescopes to } \frac{1}{2} - \frac{1}{252} = \frac{125}{252}$$

Common misconception: Cancelling terms incorrectly — writing out only the first two terms instead of enough terms to see which boundary terms survive (here it's the very first $+\frac{1}{2}$ and very last $-\frac{1}{252}$, not $\frac{1}{1}$ and $\frac{1}{251}$, since the sum starts at $\frac{1}{i+1}$ with $i = 1$).

EXAMPLE 4 — Answer

$$\sum_{i=1}^N \sum_{k=1}^i 1 = \sum_{i=1}^N i = \frac{N(N+1)}{2}; \quad \sum_{i=1}^N \sum_{k=1}^i k = \sum_{i=1}^N \frac{i(i+1)}{2} = \frac{1}{2} \left(\frac{N(N+1)(2N+1)}{6} + \frac{N(N+1)}{2} \right)$$

Common misconception: Treating the inner sum's upper limit i as if it were the constant N , e.g. writing $\sum_{k=1}^i 1 = N$ instead of i — the inner limit depends on the outer index and changes from one outer term to the next.

EXAMPLE 5 — Answer

- (a) not a partition (it's an interval, not a finite point set); (b) valid; (c) not valid (missing -1); (d) not valid (missing -1); (e) valid (as $n \rightarrow \infty$ the points approach but never reach 4 from below, and 4 is included separately, so together with the implicit... actually this set alone is infinite and has no largest element below 4 built in as a finite list — flag as **not a valid partition** since a partition must be a *finite* set of points)

Common misconception: Assuming “contains the two endpoints” is sufficient on its own without also checking that the set is finite and strictly increasing — an infinite set of points is not a partition even if it’s bounded by a and b .

EXAMPLE 6 — Answer

$L_4 = 4.146$ (approx), $R_4 = 5.146$ (approx), using $\Delta x = 1$ and $f(0) = 1, f(1) \approx 2, f(2) \approx 2.414, f(3) \approx 2.732, f(4) = 3$; R_4 overestimates since f is increasing

Common misconception: Assuming the *right*-endpoint sum is always the overestimate regardless of whether f is increasing or decreasing — it’s only the overestimate here because f is increasing; for a decreasing function the left sum overestimates.

EXAMPLE 7 — Answer

$\Delta x_1 = 1$ on $[0, 1]$, $\Delta x_2 = 2$ on $[1, 3]$; since f is decreasing, $U_P(f) = f(0)(1) + f(1)(2) = 8 + 12 = 20$, $L_P(f) = f(1)(1) + f(3)(2) = 6 + 4 = 10$

Common misconception: Using the same Δx for both subintervals (treating the partition as if it were regular) instead of computing $\Delta x_i = x_i - x_{i-1}$ separately for each piece.

EXAMPLE 8 — Answer

For increasing f : $L_P(f) = \sum_{i=1}^N f(x_{i-1})\Delta x_i$ (left endpoint gives the min), $U_P(f) = \sum_{i=1}^N f(x_i)\Delta x_i$ (right endpoint gives the max) — swapped from the decreasing case because for an increasing function the smallest value on each subinterval is at the *left* end, not the right

Common misconception: Copying the decreasing-function formulas from Page 1 without re-deriving which endpoint gives the max vs. min for an increasing function — the roles of left/right literally reverse.

EXAMPLE 9 — Answer

$$\int_0^2 x^2 dx = \lim_{n \rightarrow \infty} \frac{8}{3} \cdot (\text{ratio} \rightarrow 1) = \frac{8}{3} \quad (\text{using } \Delta x = 2/n, x_i = 2i/n, \text{ and } \sum f(x_i)\Delta x = \frac{8}{n^3} \sum i^2 \rightarrow \frac{8}{3})$$

Common misconception: Reusing the exact algebra from the worked $\int_0^1 x^2 dx$ example without redoing $\Delta x = (b - a)/n$ and x_i for the new limits $a = 0, b = 2$ — the constant out front changes from $\frac{1}{3}$ to $\frac{8}{3}$, it does not stay the same.

EXAMPLE 10 — Answer

- (a) No: if $P \subseteq Q$ we’d need $L_P(f) \leq L_Q(f)$ **and** $U_Q(f) \leq U_P(f)$ simultaneously; here $L_P = 4 \leq L_Q = 6$ ✓ but $U_Q = 9 \leq U_P = 10$ ✓ as well — so actually this data is *consistent* with $P \subseteq Q$, though it doesn’t prove it (other configurations of P, Q could produce the same numbers without one containing the other). (b) $L_{P \cup Q}(f) \geq \max(L_P, L_Q) = 6$ and $U_{P \cup Q}(f) \leq \min(U_P, U_Q) = 9$, since $P \cup Q$ refines both P and Q

Common misconception: Concluding “ $P \subseteq Q$ ” or “ $Q \subseteq P$ ” is *forced* by the numbers, when refining relationships only give one-directional inequalities — consistency with a containment is not proof of it.